

Analysis First-Year Exam, August 2023, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let (f_n) be a sequence of continuous functions from $[0, 1]$ to $[0, 1]$. For each of the following, give a proof (if true) or a counterexample (if false):
 - (i) if $f_n(t)$ decreases pointwise to 0 then $\int_0^1 f_n(t) dt \rightarrow 0$;
 - (ii) if $f_n(t)$ only converges pointwise to 0 then $\int_0^1 f_n(t) dt \rightarrow 0$.
2. For $0 < t < 1$ and for n a positive integer, let $f_n(t) = t^n$.
 - (i) Show that the sequence (f_n) converges uniformly on each compact subset of $(0, 1)$.
 - (ii) Show that (f_n) does not converge uniformly on $(0, 1)$.
3. Let the function $f : [1, \infty) \rightarrow \mathbb{R}$ be continuous, with $\lim_{t \rightarrow \infty} f(t) = A \in \mathbb{R}$. Prove there exists a sequence (p_n) of polynomials such that $p_n(1/t) \rightarrow f(t)$ uniformly for $t \geq 1$.
4. Let $A \subseteq [0, 1]$ have positive Lebesgue measure: that is, $\lambda(A) > 0$. Prove that if $0 < \alpha < \lambda(A)$ then there exists a Lebesgue measurable set $A_\alpha \subseteq A$ such that $\lambda(A_\alpha) = \alpha$.
5. State the Monotone Convergence Theorem and ‘Fatou’s Lemma’. Deduce one of these from the other.